Classification of States

Definition. Accessibility

State j is **accessible** from state i, denoted by $i \to j$, if

$$p_{ij}^{(n)} > 0 \quad \text{for some } n \geq 0.$$

Remark.

- Accessibility is not a one-step concept: p_{ij} may be zero while $p_{ij}^{(n)} > 0$ for some $n > 1$.
- It is not symmetric: $i \to j$ does not imply $j \to i$.
- Since $p_{ii}^{(0)} = 1$, every state is accessible from itself.

Definition. Communication

States i and j **communicate**, denoted by $i \leftrightarrow j$, if

$$i \to j \qquad \text{and} \qquad j \to i.$$

Note. Communication is an **equivalence** relation.

Definition. Communication Class

A **communication class** is a maximal set $\mathscr{C} \subseteq S$ whose states communicate with one another.

Note. Communication classes form a partition of the state space S.

Definition. Irreducible chains

A Markov chain is **irreducible** if it has exactly one communication class. Otherwise, it is **reducible**.

Definition. Closed Class

A communication class $\mathscr{C}$ is **closed** if the chain cannot leave it:

$$p_{ij} = 0 \qquad \text{for every } i \in \mathscr{C} \text{ and } j \notin \mathscr{C}.$$

Remark.

- Once $X_n \in \mathscr{C}$ for any n, all future states X_{n+m} remain in $\mathscr{C}$ with probability 1, any $m \geq 0$.
- An **absorbing state** i satisfies $p_{ii} = 1$. Hence $\{i\}$ is a closed class.
- A closed class may contain several states and the chain may continue moving inside it.

Theorem

Every finite-state Markov chain has at least one closed communicating class.

Definition. Hitting and First-Return Times

For a state j, define

$$T_j = \inf\{n \geq 0 : X_n = j\}.$$

For a process **starting from state** i, define the **first *return* time**

$$T_i^+ = \inf\{n \geq 1 : X_n = i\}.$$

Remark.

- T_j: the first time to arrive at state j.
- T_i^+: we know that we start from i, and we wonder the first return time to i.

Definition. First-Return Probability

Starting from i, the probability that the first return occurs exactly at time n is denoted as

$$f_{ii}^{(n)} = P_i(T_i^+ = n), \qquad n \geq 1.$$

The probability of ever returning is

$$f_{ii} = P_i(T_i^+ < \infty) = \sum_{n=1}^{\infty} f_{ii}^{(n)}.$$

Takeaway. Consider state i. If the process returns to i almost surely (with prob. 1), i.e. $f_{ii} = 1$, then it will return to i infinitely many times, and the total returning prob. $\sum_{n=1}^{\infty} p_{ii}^{(n)}$ is infinite. We call i **recurrent**. This is a class property.

Definition. Recurrence and Transience

A state i is **recurrent** if

$$P_i(T_i^+ < \infty) = 1.$$

It is **transient** if

$$P_i(T_i^+ < \infty) < 1.$$

Define the number of visits to i after time zero:

$$N_i = \sum_{n=1}^{\infty} \mathbf{1}\{X_n = i\}.$$

Theorem. Number of Revisits

For every $m \geq 1$,

$$P_i(N_i \geq m) = (f_{ii})^m.$$

Consequently,

$$\begin{cases} N_i = \infty \text{ a.s.}, & f_{ii} = 1, \\ N_i < \infty \text{ a.s.}, & f_{ii} < 1. \end{cases}$$

$$E_i[N_i] = \begin{cases} \dfrac{f_{ii}}{1 - f_{ii}}, & f_{ii} < 1, \\ \infty, & f_{ii} = 1. \end{cases}$$

Hence,

$$i \text{ recurrent} \iff N_i = \infty \text{ a.s.} \iff E_i[N_i] = \infty,$$

$$i \text{ transient} \iff N_i < \infty \text{ a.s.} \iff E_i[N_i] = \frac{f_{ii}}{1 - f_{ii}}.$$

Define the n-step return probability

$$p_{ii}^{(n)} = P_i(X_n = i).$$

Theorem. Return-Probability Criterion

$$i \text{ is recurrent} \iff \sum_{n=1}^{\infty} p_{ii}^{(n)} = \infty,$$

$$i \text{ is transient} \iff \sum_{n=1}^{\infty} p_{ii}^{(n)} < \infty.$$

Starting from state i,

$$\begin{aligned} i \text{ recurrent} &\iff P_i(T_i^+ < \infty) = f_{ii} = 1 \\ &\iff P_i(N_i = \infty) = 1 \\ &\iff E_i[N_i] = \infty \\ &\iff \sum_{n=1}^{\infty} p_{ii}^{(n)} = \infty. \end{aligned}$$

Remark.

- $f_{ii}^{(n)} = P_i(T_i^+ = n)$ counts paths whose *first* return occurs at time n.
- $p_{ii}^{(n)} = P_i(X_n = i)$ counts all paths that are at i at time n, including paths that returned earlier.

Note. So, $f_{ii}^{(n)} \leq p_{ii}^{(n)}$.

Theorem

If $i \leftrightarrow j$, then i and j are either both recurrent or both transient.

Theorem

For a **finite-state** Markov chain:

- at least **one closed communicating class** exists;
- recurrent class $\iff$ closed class;
- all states in a nonclosed class are transient.

Definition. Period

For a state i, define

$$d(i) = \gcd\{n \geq 1 : p_{ii}^{(n)} > 0\}.$$

If the set is empty, use the convention $d(i) = 0$.
A state is **aperiodic** if $d(i) = 1$.

Theorem

If $i \leftrightarrow j$, then

$$d(i) = d(j).$$

Definition. Aperiodic Chain

A Markov chain is **aperiodic** if every state is aperiodic, i.e. $d(i) = 1$ for all $i \in S$.

Theorem

Let state i have period $d = d(i) \geq 1$.

- For every $n \geq 1$,

$$d \nmid n \quad \Longrightarrow \quad p_{ii}^{(n)} = 0.$$

- There exists an integer $N_i \geq 1$ such that

$$p_{ii}^{(kd)} > 0 \qquad \text{for every integer } k \geq N_i.$$

Note. Some small multiples of d may not be possible return times.

Definition. Regularity

A **finite-state** Markov chain with transition matrix $\mathbf{P}$ is **regular** if there exists an integer $r \geq 1$ such that **every entry of $\mathbf{P}^r$ is strictly positive**; that is,

$$p_{ij}^{(r)} > 0 \qquad \text{for all } i, j \in S.$$

Theorem

For a finite-state Markov chain,

regular $\iff$ irreducible and aperiodic.

Theorem

Let $\{X_n\}$ be a **regular** Markov chain on a finite state space S, with transition matrix $\mathbf{P}$. There exists a distribution $\boldsymbol{\pi} = (\pi(j))_{j\in S}$ such that, for every initial distribution $\boldsymbol{\pi}_0$,

$$\lim_{n\to\infty} \boldsymbol{\pi}_n = \lim_{n\to\infty} \boldsymbol{\pi}_0 \mathbf{P}^n = \boldsymbol{\pi}.$$

- The convergence is entrywise convergence, i.e. $\pi_n(j) \to \pi(j)$ for all $j \in S$.
- This limit is independent of $\boldsymbol{\pi}_0$, i.e. $p_{ij}^{(n)} \to \pi(j)$ for all $i, j \in S$.
- $\boldsymbol{\pi}$ is unique, and it satisfies

$$\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}, \qquad \sum_{j\in S} \pi(j) = 1, \qquad \pi(j) > 0.$$

To calculate the limiting distribution, solve

$$\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}, \qquad \sum_{j\in S} \pi(j) = 1, \qquad \pi(j) \geq 0.$$

- Before calculation, **verify the convergence first!**
- This is a **linear system** in the unknowns $(\pi(j))_{j\in S}$.

Remark. The jth entry $\pi(j)$ of $\boldsymbol{\pi}$ can be interpreted as the **long-run fraction of time spent in state** j:

$$\left(\frac{1}{n}\sum_{k=1}^{n} \mathbf{1}_{\{X_k = j\}}\right)_{j\in S} \xrightarrow{\text{a.s.}} \boldsymbol{\pi}.$$

This convergence holds with probability 1.

Definition. Positive and Null Recurrence

For a recurrent state j, let $m_j = \mathbb{E}_j[T_j^+]$, where T_j^+ is the first return time to state j.

- **Positive** recurrent: $m_j < \infty$.
- **Null** recurrent: $m_j = \infty$.

Remark.

- Recurrence guarantees a return, but the mean may be infinite. Positive recurrence further guarantees a finite return time.
- Positive and null recurrence are class properties.
- Every finite irreducible chain is positive recurrent.
- Null recurrence only applies for infinite state MCs. For finite state MCs, any recurrent state is positive recurrent.

Definition. Ergodicity

A Markov chain is **ergodic** if it is irreducible, **positive recurrent**, and aperiodic.

Theorem. Limiting Distribution

An ergodic chain on a finite or countable state space S has a unique limiting distribution $\boldsymbol{\pi} = (\pi(j))_{j\in S}$. For every initial distribution $\boldsymbol{\pi}_0$,

$$\lim_{n\to\infty} \pi_n(j) = \pi(j) = \frac{1}{m_j} > 0, \qquad j \in S.$$

- The limit is independent of $\boldsymbol{\pi}_0$.
- It satisfies

$$\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}, \qquad \sum_{j\in S} \pi(j) = 1.$$

- For finite chains, ergodicity is equivalent to regularity.
- Irreducible + aperiodic + finite states ⇒ limiting dist.
- Irreducible + aperiodic + positive recurrent ⇒ limiting dist.
- **Convergence.** Finite chains: regular, equivalently irreducible and aperiodic. Countable chains: irreducible, positive recurrent and aperiodic. Then $\boldsymbol{\pi}_n \to \boldsymbol{\pi}$, independently of $\boldsymbol{\pi}_0$.
- **Calculation.** Verify convergence first; then solve $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$, $\sum_j \pi(j) = 1$, $\pi(j) \geq 0$. For an ergodic chain, $\pi(j) = 1/m_j$.
- **Interpretation.** In an ergodic chain, fixed-time probabilities and one-path visit frequencies both approach $\pi(j)$.
- **Periodicity and null recurrence.** Positive recurrent periodic chains retain phase limits and time averages, including one-path frequencies. For a null recurrent state j, $\pi_n(j) \to 0$.
- **Multiple classes.** Combine entering probabilities with within-class behaviour. The starting state can affect the limit.

	Transient	Null recurrent	Positive recurrent
$\mathbb{P}_i(T_i^+ < \infty)$	$f_{ii} < 1$	1	1
N_i	finite a.s.	∞ a.s.	∞ a.s.
$m_i = \mathbb{E}_i[T_i^+]$	∞	∞	finite
$\mathbb{E}_i[N_i]$	$f_{ii}/(1-f_{ii})$	∞	∞
Long-run fraction	0	0	$1/m_i > 0$

Shared Properties Within a Class

Property	Shared by all states in the class?	Explanation
Recurrence or transience	Yes	If one state is recurrent, every state in the class is recurrent. Otherwise, they are all transient.
Positive or null recurrence	Yes	All recurrent states in the class are either positive recurrent or null recurrent.
Period	Yes	If one state has period d, every communicating state has the same period d.
Closedness	Yes, as a class property	A communicating class is either closed or not closed. If one state has a possible transition outside the class, the class is not closed.
Communication structure	Yes	By definition, every pair of states in the class communicates.

Properties that are not necessarily shared. The following **values or detailed behaviors can differ from state to state**:

- The expected first-return time;
- The stationary probability $\pi(i)$;
- The probability of returning at a particular time n;
- Whether there is a one-step self-loop;
- The exact set of possible return times;
- Transition probabilities to individual states;
- Expected hitting times to another state.

Important distinction about period. The **period** is shared, but the exact return-time probabilities need not be.

For example, two states may both have period 3, meaning their possible return times have greatest common divisor 3. However, one state might be able to return after 3 steps while another can first return only after 6 steps. The period is still the same.

Stationary Distributions

Definition. Stationary Distribution

A probability vector $\boldsymbol{\pi} = (\pi(j))_{j\in S}$ is a **stationary distribution** for $\mathbf{P}$ if $\boldsymbol{\pi} = \boldsymbol{\pi}\mathbf{P}$.

Definition. Global Balance Equations

A probability vector $\boldsymbol{\pi}$ satisfies **global balance** if, for every j,

$$\underbrace{\pi(j)}_{\text{Leaving prob.}} = \underbrace{\sum_i \pi(i) p_{ij}}_{\text{Arriving prob.}}.$$

Equivalently, excluding self-transitions, $\pi(j)\sum_{k\neq j} p_{jk} = \sum_{i\neq j} \pi(i) p_{ij}$.

Definition. Detailed Balance Equations

A probability vector $\boldsymbol{\pi}$ satisfies the **detailed balance equations** if, for every pair i, j,

$$\pi(i) p_{ij} = \pi(j) p_{ji}.$$

Note. Detailed balance ⇒ global balance, but the converse is not true.

Theorem

For an **irreducible** chain, the existence of a stationary probability distribution implies **positive recurrence**.